8.5 Integration of Rational Functions by Partial Fractions

The method for rewriting rational functions as a sum of simpler fractions is called **the method of partial fractions**. For instant, we can express $\frac{5}{x^2-1}$ as $\frac{A}{x-1}+\frac{B}{x+1}$. We call the fractions $\frac{A}{x-1}$ and $\frac{B}{x+1}$ **partial fractions** because their denominators are only part of the original denominator x^2-1. We call A and B **undetermined coefficients** until suitable values for them have been found.

General Description of the Method

Success in writing a rational function $f(x)/g(x)$ as a sum of partial fractions depends on three things:

- *The degree of* $f(x)$ *must be less than the degree of* $g(x)$. That is, the fraction must be proper. proper. If it isn't, divide $f(x)$ by $g(x)$ and work with the remainder term.
- We must know the factors of $g(x)$. In theory, any polynomial with real coefficients can be written as a product of real linear factors and real quadratic factors. In practice, the factors may be hard to find.
- The values of the undetermined coefficients form a system of n linear equations in n unknowns. For large n, solving such systems may require linear algebra methods (such as Gaussian Elimination).

Here is how we find the partial fractions of a proper fraction $f(x)/g(x)$ when the factors of g are known. A quadratic polynomial (or factor) is **irreducible** if it cannot be written as the product of two linear factors with real coefficients. That is, the polynomial has no real roots.

Method of Partial Fractions When $f(x)/g(x)$ Is Proper

1. Let $x-r$ be a linear factor of $g(x)$. Suppose that $(x-r)^m$ is the highest power of $x-r$ that divides $g(x)$. Then, to this factor, assign the sum of the m partial fractions:
$$\frac{A_1}{x-r}+\frac{A_2}{(x-r)^2}+\cdots+\frac{A_m}{(x-r)^m}.$$
Do this for each distinct linear factor of g(x).
2. Let x^2+px+q be an **irreducible quadratic** factor of $g(x)$. In this case, x^2+px+q has no real roots. Suppose that $(x^2+px+q)^n$ is the highest power of this factor that divides $g(x)$. Then, to this factor, assign the sum of the n partial fractions:
$$\frac{B_1x+C_1}{x^2+px+q}+\frac{B_2x+C_2}{(x^2+px+q)^2}+\cdots+\frac{B_nx+C_n}{(x^2+px+q)^n}.$$
Do this for each distinct quadratic factor of $g(x)$.
3. Set the original fraction $f(x)/g(x)$ equal to the sum of all these partial fractions. Clear the resulting equation of fractions.
4. Find the values of the undetermined coefficients.

There are often multiple ways to find the values of the undetermined coefficients in Step 4. To find the values of the undetermined coefficients, we equated coefficients of like powers of x. We also can assign convenient values of x, leading to simple equations that we can solve for the undetermined coefficients.

Example 1: Evaluate $\int \frac{x^2+4x+1}{(x-1)(x+1)(x+3)}\,dx.$

Solution:
Note that each of the factors $x - 1$, $x + 1$, and $x + 3$ is raised only to the first power. Therefore, the partial fraction decomposition has the form

$$\frac{x^2 + 4x + 1}{(x - 1)(x + 1)(x + 3)} = \frac{A}{x - 1} + \frac{B}{x + 1} + \frac{C}{x + 3}$$

To find the values of the undetermined coefficients A, B, and C, we clear fractions and get

$$x^2 + 4x + 1 = A(x + 1)(x + 3) + B(x - 1)(x + 3) + C(x - 1)(x + 1).$$

Let $x = 1$: $1^2 + 4(1) + 1 = A(2)(4) + B(0) + C(0) \Rightarrow 6 = 8A \Rightarrow A = 3/4.$

Let $x = -1$: $(-1)^2 + 4(-1) + 1 = A(0) + B(-4) + C(0) \Rightarrow -2 = -4B \Rightarrow B = 1/2.$

Let $x = -3$: $(-3)^2 + 4(-3) + 1 = A(0) + B(0) + C(8) \Rightarrow -2 = 8C \Rightarrow C = -1/4.$

Hence, $\int \frac{x^2+4x+1}{(x-1)(x+1)(x+3)}\,dx = \int \frac{3dx}{4(x-1)} + \int \frac{dx}{2(x+1)} - \int \frac{dx}{4(x+3)}$

$$= \frac{3}{4}\ln|x - 1| + \frac{1}{2}\ln|x + 1| - \frac{1}{4}\ln|x + 3| + K.$$

where K is the arbitrary constant of integration (we call it K here to avoid confusion with the undetermined coefficient we labelled as C).

You can solve for the undetermined coefficients (A, B, etc.) by equating coefficients of like powers of x or by assigning convenient values to x. You should choose the method that is most convenient for the problem at hand.

Example 2: Evaluate $\int \frac{6x+7}{(x+2)^2}\,dx.$

Solution:

$$\frac{6x + 7}{(x + 2)^2} = \frac{A}{x + 2} + \frac{B}{(x + 2)^2}$$

To find the values of the undetermined coefficients A, B, and C, we clear fractions and get

$$6x + 7 = A(x + 2) + B = Ax + 2A + B.$$

Equating coefficients of corresponding powers of x gives

$A = 6$ and $2A + B = 2(6) + B = 7 \Rightarrow$ $A = 6$ and $B = -5$.

Hence, $\int \frac{6x+7}{(x+2)^2}\,dx = \int \frac{6}{x+2}\,dx + \int \frac{-5}{(x+2)^2}\,dx = 6\ln|x + 2| + \frac{5}{x+2} + C.$

Example 3: Evaluate $$\int \frac{2x^3 - 4x^2 - x - 3}{x^2 - 2x - 3}\,dx.$$

Solution First we divide the denominator into the numerator to get a polynomial plus a proper fraction.

$$\begin{array}{r} 2x \\ x^2 - 2x - 3\overline{)\,2x^3 - 4x^2 - x - 3} \\ \underline{2x^3 - 4x^2 - 6x - 3} \\ 5x - 3 \end{array}$$

Then we write the improper fraction as a polynomial plus a proper fraction.

$$\frac{2x^3 - 4x^2 - x - 3}{x^2 - 2x - 3} = 2x + \frac{5x - 3}{x^2 - 2x - 3}$$

We found the partial fraction decomposition of the fraction on the right in the opening example, so

$$\begin{aligned} \int \frac{2x^3 - 4x^2 - x - 3}{x^2 - 2x - 3}\,dx &= \int 2x\,dx + \int \frac{5x - 3}{x^2 - 2x - 3}\,dx \\ &= \int 2x\,dx + \int \frac{2}{x + 1}\,dx + \int \frac{3}{x - 3}\,dx \\ &= x^2 + 2\ln|x + 1| + 3\ln|x - 3| + C. \end{aligned}$$ ■

Example 4: Evaluate $$\int \frac{-2x + 4}{(x^2 + 1)(x - 1)^2}\,dx.$$

Solution The denominator has an irreducible quadratic factor $x^2 + 1$ as well as a repeated linear factor $(x - 1)^2$, so we write

$$\frac{-2x + 4}{(x^2 + 1)(x - 1)^2} = \frac{Ax + B}{x^2 + 1} + \frac{C}{x - 1} + \frac{D}{(x - 1)^2}. \tag{2}$$

Clearing the equation of fractions gives

$$\begin{aligned} -2x + 4 &= (Ax + B)(x - 1)^2 + C(x - 1)(x^2 + 1) + D(x^2 + 1) \\ &= (A + C)x^3 + (-2A + B - C + D)x^2 \\ &\quad + (A - 2B + C)x + (B - C + D). \end{aligned}$$

Equating coefficients of like terms gives

$$\begin{aligned} \text{Coefficients of } x^3: &\quad 0 = A + C \\ \text{Coefficients of } x^2: &\quad 0 = -2A + B - C + D \\ \text{Coefficients of } x^1: &\quad -2 = A - 2B + C \\ \text{Coefficients of } x^0: &\quad 4 = B - C + D \end{aligned}$$

We solve these equations simultaneously to find the values of A, B, C, and D:

$$\begin{aligned} -4 &= -2A, \qquad A = 2 && \text{Subtract fourth equation from second.} \\ C &= -A = -2 && \text{From the first equation} \\ B &= (A + C + 2)/2 = 1 && \text{From the third equation and } C = -A \\ D &= 4 - B + C = 1. && \text{From the fourth equation} \end{aligned}$$

We substitute these values into Equation (2), obtaining

$$\frac{-2x + 4}{(x^2 + 1)(x - 1)^2} = \frac{2x + 1}{x^2 + 1} - \frac{2}{x - 1} + \frac{1}{(x - 1)^2}.$$

Finally, using the expansion above we can integrate:

$$\begin{aligned} \int \frac{-2x + 4}{(x^2 + 1)(x - 1)^2}\,dx &= \int \left(\frac{2x + 1}{x^2 + 1} - \frac{2}{x - 1} + \frac{1}{(x - 1)^2} \right) dx \\ &= \int \left(\frac{2x}{x^2 + 1} + \frac{1}{x^2 + 1} - \frac{2}{x - 1} + \frac{1}{(x - 1)^2} \right) dx \\ &= \ln (x^2 + 1) + \tan^{-1} x - 2 \ln |x - 1| - \frac{1}{x - 1} + C. \end{aligned}$$

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Example 5: Evaluate

$$\int \frac{dx}{x(x^2 + 1)^2}.$$

Solution The form of the partial fraction decomposition is

$$\frac{1}{x(x^2 + 1)^2} = \frac{A}{x} + \frac{Bx + C}{x^2 + 1} + \frac{Dx + E}{(x^2 + 1)^2}.$$

Multiplying by $x(x^2 + 1)^2$, we have

$$\begin{aligned} 1 &= A(x^2 + 1)^2 + (Bx + C)x(x^2 + 1) + (Dx + E)x \\ &= A(x^4 + 2x^2 + 1) + B(x^4 + x^2) + C(x^3 + x) + Dx^2 + Ex \\ &= (A + B)x^4 + Cx^3 + (2A + B + D)x^2 + (C + E)x + A. \end{aligned}$$

If we equate coefficients, we get the system

$$A + B = 0, \qquad C = 0, \qquad 2A + B + D = 0, \qquad C + E = 0, \qquad A = 1.$$

Solving this system gives $A = 1$, $B = -1$, $C = 0$, $D = -1$, and $E = 0$. Thus,

$$\int \frac{dx}{x(x^2+1)^2} = \int \left[\frac{1}{x} + \frac{-x}{x^2+1} + \frac{-x}{(x^2+1)^2}\right] dx$$

$$= \int \frac{dx}{x} - \int \frac{x\,dx}{x^2+1} - \int \frac{x\,dx}{(x^2+1)^2}$$

$$= \int \frac{dx}{x} - \frac{1}{2}\int \frac{du}{u} - \frac{1}{2}\int \frac{du}{u^2} \qquad u = x^2+1,\ du = 2x\,dx$$

$$= \ln|x| - \frac{1}{2}\ln|u| + \frac{1}{2u} + K$$

$$= \ln|x| - \frac{1}{2}\ln(x^2+1) + \frac{1}{2(x^2+1)} + K$$

$$= \ln\frac{|x|}{\sqrt{x^2+1}} + \frac{1}{2(x^2+1)} + K.$$

Determining Coefficients by Differentiating

Another way to determine the constants that appear in partial fractions is to differentiate, as in the next example.

Example 6: Find A, B, and C in the equation

$$\frac{x-1}{(x+1)^3} = \frac{A}{x+1} + \frac{B}{(x+1)^2} + \frac{C}{(x+1)^3}$$

by clearing fractions, differentiating the result, and substituting $x = -1$.

Solution We first clear fractions:

$$x - 1 = A(x+1)^2 + B(x+1) + C.$$

Substituting $x = -1$ shows $C = -2$. We then differentiate both sides with respect to x, obtaining

$$1 = 2A(x+1) + B.$$

Substituting $x = -1$ shows $B = 1$. We differentiate again to get $0 = 2A$, which shows $A = 0$. Hence,

$$\frac{x-1}{(x+1)^3} = \frac{1}{(x+1)^2} - \frac{2}{(x+1)^3}.$$

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In some problems, assigning small values to x, such as $x = 0, \pm 1, \pm 2$, to get equations in A, B, and C provides a fast alternative to other methods.